

Generic CHSH Violation in Fibonacci Anyon Braiding: A Landscape Analysis

Berkay Yüksel Sayim^{1, 2, *}

¹*Independent Research, Germany*

²*ORCID: 0009-0004-4993-7352*

(Dated: 2026-07-23)

We report the first, to our knowledge, complete sequence-by-sequence numerical landscape of CHSH-inequality violation across all Fibonacci anyon braiding words of length $L = 3\text{--}12$ in two complementary encodings of six anyons. In the two-generator d_{1b} encoding (native four-dimensional fusion space), the optimal sequence at $L = 12$ reaches $|S| = 2.811$ (99.4% of the Tsirelson bound $2\sqrt{2}$) with concurrence $C = 0.988$. Throughout, “CHSH violation” denotes a CHSH value exceeding 2 on the Fibonacci fusion space, which does not factorize into spatially separated Alice/Bob subsystems; the reported values are signatures of topological nonseparability and single-algebra contextuality, not Bell violations in the Einstein–Podolsky–Rosen sense. In the three-generator encoding under four-dimensional computational-subspace projection, the enumeration maximum approaches the Tsirelson bound to within 0.16% already at $L = 8$. CHSH violation is *generic*: 64.6% of sequences violate the CHSH inequality at $L = 9$ under native five-dimensional measurement, rising to 91.8% under computational-subspace projection. A single topological mechanism controls this landscape: blocking the cross-bipartition generator σ_3 collapses concurrence to $C = 0$ exactly for all sequences and all sector phases—topology acts as a gatekeeper for entanglement. Sequential braiding on a shared fusion space produces positive lag-1 autocorrelation $\text{ACF}(1) = +0.34$ at 10.85σ significance, a falsifiable state-continuity signature not predicted by standard quantum mechanics under independent measurement settings. Together with a companion mutual-information analysis of topological sectors and CHSH values, these results indicate that the available topological operations gate whether CHSH correlations are accessible on the fusion space.

I. INTRODUCTION

Non-Abelian anyons in two-dimensional topological phases support braiding operations that can generate entanglement between logical qubits encoded in the fusion space [3, 4]. For Fibonacci anyons—the simplest non-Abelian species with a universal gate set—individual braiding sequences that violate the CHSH inequality have been identified in several contexts [5, 6]. However, a basic question has remained open: *what fraction of all possible braiding sequences produces CHSH violation, and what controls the boundary between violating and nonviolating sequences?*

We address this question by exhaustive enumeration. For six Fibonacci anyons on a linear fusion tree, we compute the maximal CHSH value $|S|$ for every braiding sequence in two distinct encodings: a two-generator encoding (d_{1b} , $L = 3\text{--}12$, native four-dimensional Hilbert space) and a three-generator encoding ($L = 2\text{--}8$ fully enumerated, $L = 9$ for CHSH-fraction statistics; full five-dimensional fusion space). The enumeration comprises 2^L sequences in d_{1b} and 3^L sequences in the three-generator case (up to $3^9 = 19683$), totaling over 10^6 evaluated unitaries.

Three principal findings emerge:

(i) *CHSH violation is generic.* A majority of sequences violate the CHSH inequality under both mea-

surement protocols we consider—64.6% in the native five-dimensional protocol and 91.8% under computational-subspace projection at $L = 9$.

(ii) *A single generator acts as a topological switch.* In the three-generator encoding, removing the cross-bipartition generator σ_3 from the gate set collapses the concurrence to $C = 0$ exactly, for all sequences and all topological sector phases; the two-generator d_{1b} encoding contains no σ_3 , and its entanglement instead arises indirectly through the shared mediator (Sec. IID).

(iii) *Sequential braiding produces temporal correlations.* When braiding operations are applied sequentially to a shared anyon register, the resulting CHSH-value time series exhibits positive autocorrelation ($\text{ACF}(1) = +0.34$, significance 10.85σ), while a null test with independent sector phases yields $\text{ACF} \approx 0$.

These results complement the mutual-information study between topological sectors and CHSH values reported in Ref. [1], which found that classical lattice models ($\mathbb{Z}_2$ gauge theory, XY model) exhibit finite-size correlations $I(T; S) > 0$ between sector labels and Bell-type observables (a finite-size effect that a scaling analysis indicates decays toward zero with system size).

The present work differs from prior studies of Fibonacci braiding in scope and focus. Burke *et al.* [9] characterized the Weyl-chamber structure of Fibonacci braiding gates but did not evaluate CHSH violation. Cui *et al.* [8] established that exact leakage-free entangling gates may not exist for Fibonacci anyons, motivating our two-protocol analysis. Yang *et al.* [10] studied information scrambling through braiding but not CHSH cor-

* berkya@tutamail.com

relations. Experimental demonstrations of non-Abelian statistics [11] have confirmed braiding phases but have not mapped the CHSH landscape. While prior work by Brennen *et al.* [15] established Bell-test constructions for Fibonacci/Ising anyons, Chang, Chu, and Ma [16] studied Bell’s inequality in topologically ordered qubit models (Wen-plaquette), and Xu, Ye, and You [20] have independently and contemporaneously analyzed three-copy CHSH correlations on Fibonacci anyonic states with a fusion-sector observables construction, to our knowledge no prior work presents the complete sequence-by-sequence enumeration of CHSH violation fractions over all braid words of a given length class — the contribution of the present paper. Earlier qualitative Bell tests on Majorana-based topological platforms appear in Drummond *et al.* [17] and Romito and Gefen [18], with the Clifford-only limitation of Ising braiding analyzed by Clarke, Sau, and Das Sarma [19].

The paper is organized as follows. Section II defines the two encodings and the CHSH evaluation method. Section III presents the enumeration results and reconciles the two encodings. Section IV establishes the σ_3 -blocking result. Section V analyzes the measurement-protocol dependence of CHSH fractions. Section VI presents the sequential-braiding autocorrelation. Section VII discusses implications and limitations.

II. FRAMEWORK

A. Fusion space of six Fibonacci anyons

We consider six Fibonacci anyons $\tau_1, \dots, \tau_6$ arranged on a linear fusion tree with total charge constrained to the vacuum sector. The Fibonacci fusion rule $\tau \times \tau = \mathbf{1} \oplus \tau$ gives a five-dimensional fusion space spanned by the intermediate charge labels (x_1, x_2, x_3) , where x_k is the total charge of the first $2k$ anyons:

$$\begin{aligned} |0\rangle &= (1, \tau, 1), & |1\rangle &= (1, \tau, \tau), \\ |2\rangle &= (\tau, 1, \tau), & |3\rangle &= (\tau, \tau, 1), \\ |4\rangle &= (\tau, \tau, \tau). \end{aligned} \quad (1)$$

The fourth intermediate label $x_4 = d$ is fixed to τ by the vacuum constraint.

B. F- and R-matrices

The Fibonacci F -matrix in the $(\mathbf{1}, \tau)$ basis is

$$F = \begin{pmatrix} \varphi^{-1} & \varphi^{-1/2} \\ \varphi^{-1/2} & -\varphi^{-1} \end{pmatrix}, \quad \varphi = \frac{1 + \sqrt{5}}{2}, \quad (2)$$

satisfying $F^2 = I$ to machine precision ($\Delta < 10^{-17}$). The R -eigenvalues for the two fusion channels are

$$R_1 = e^{-4\pi i/5}, \quad R_\tau = e^{+3\pi i/5}. \quad (3)$$

C. Braid generators

The five braid generators $\sigma_1, \dots, \sigma_5$ act as 5×5 unitary matrices on the fusion-space basis (1). Each σ_k exchanges anyons τ_k and τ_{k+1} via the sequence $F^{-1} \cdot \text{diag}(R_1, R_\tau) \cdot F$ on the relevant two-dimensional subspace, with the remaining basis states acquiring diagonal R -phases. Explicit matrix forms are given in Appendix A.

All four Yang-Baxter relations $\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$ are satisfied with residuals $\Delta \leq 3.3 \times 10^{-16}$, and distant generators commute ($\Delta \leq 1.5 \times 10^{-17}$ for $|i - j| \geq 2$) [1].

D. Two encodings

We study two encodings of logical qubits into the fusion space:

d_{1b} encoding (two generators). The logical qubits are $Q_A = x_1$ (Alice) and $Q_B = x_3$ (Bob), with the mediator $x_2 = m(x_1, x_3)$ uniquely determined by the vacuum constraint. This yields a four-dimensional Hilbert space $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ with no leakage. The available gates are $\text{AM} = \sigma_2$ (braids $\tau_2 \tau_3$, acts on Alice) and $\text{MB} = \sigma_4$ (braids $\tau_4 \tau_5$, acts on Bob). Neither gate directly crosses the Alice–Bob bipartition; entanglement arises indirectly through the shared mediator m , as evidenced by the nonvanishing commutator $\|[\text{AM}, \text{MB}]\|_\infty \approx 0.30$.

A topological sector phase δ is introduced by modifying $R_\tau \rightarrow R_\tau \cdot e^{i\delta}$, parameterizing the family of braid representations on the torus.

Three-generator encoding (full 5D). All five basis states (1) are retained, and the gate set is $\{\sigma_2, \sigma_3, \sigma_4\}$. Here σ_2 acts locally on Alice (x_1), σ_4 acts locally on Bob (x_3), and σ_3 directly crosses the bipartition by refusing x_2 . This provides the standard structure of single-qubit rotations plus entangling gate for universal quantum computation [5].

E. Interpretation: nonseparability and contextuality

Throughout this paper, “CHSH violation” refers to CHSH values exceeding 2 computed on the Fibonacci anyon fusion space. This fusion space does not admit a tensor-product factorization into spatially separated Alice/Bob subsystems: the effective measurement operators in the d_{1b} encoding fail to commute ($\|[\text{AM}, \text{MB}]\|_\infty \approx 0.30$, Sec. IID). Accordingly, the reported values are signatures of *topological nonseparability* on a nonfactorized fusion space, not Bell violations in the Einstein–Podolsky–Rosen sense.

This noncommutativity is a property of the specific d_{1b} projection, not of the underlying full algebra: on the full five-dimensional fusion space, the Protocol B observables commute exactly across the bipartition ($[A_i, B_j] = 0$ for all i, j , Sec. V), so Tsirelson’s bound remains the relevant

ceiling and is respected throughout (at most $|S| = 2.824$, 99.84% of $2\sqrt{2}$, here). We therefore classify the reported correlations as a form of *topological, single-algebra contextuality* [24] rather than EPR-type nonlocality, in the spirit of algebraic-quantum-field-theory results showing that maximal Bell-CHSH violation is generic for correlations defined on a single non-Abelian algebra, without requiring spacelike separation [39], and of related studies of Bell-operator representations on single quantum systems [23, 38]. This reading is consistent with the anyonic-entanglement structure described in Ref. [25], and with the broader point—sharpened by the resolution of the Tsirelson problem [40]—that the appropriate correlation bound depends on the algebraic separation assumed between the measurement structures, not merely on the numerical value of $|S|$. Blanket no-go claims against such single-algebra CHSH values are folklore rather than sharp theorems in this specific setting [41]; we make no prohibition claim here and instead report the values as topologically gated, nonseparable correlations bounded by, but not violating, Tsirelson’s inequality.

F. CHSH evaluation

For each braiding sequence $\mathcal{B}$ and sector phase δ , we compute the unitary $U(\mathcal{B}, \delta)$ and the output state $|\psi\rangle = U|0\rangle$. The maximal CHSH value is obtained via the Horodecki criterion [7]: for a pure two-qubit state with concurrence C ,

$$|S|_{\max} = 2\sqrt{1 + C^2}. \quad (4)$$

This bound is tight and achieved by optimal measurement angles derived from the singular-value decomposition of the correlation matrix $T_{ij} = \text{Re} \langle \psi | A_i \otimes B_j | \psi \rangle$.

In the d_{1b} encoding, C is the concurrence of the 4×4 density matrix ρ_{AB} . In the three-generator encoding, we evaluate CHSH under two protocols described in Sec. V.

For each length L , the CHSH-violation fraction is defined as $\text{CHSH}\% = |\{(\mathcal{B}, \delta) : |S| > 2\}| / |\{(\mathcal{B}, \delta)\}|$.

III. RECONCILIATION OF TWO ENCODINGS

A. d_{1b} : Two-generator landscape

Table I presents the optimal CHSH value, concurrence, and winning sequence for each braid length $L = 3$ –12 in the d_{1b} encoding, obtained by exhaustive enumeration over all 2^L sequences; the per-sequence sector-phase optima are reported on the 50-point grid of the deposited `s_delta_sweeps.json`. (At $L = 2$, a single gate yields $|S| = 2.097$, $C = 0.315$.)

The concurrence increases monotonically from $C = 0.457$ at $L = 3$ to $C = 0.988$ at $L = 12$, with no saturation plateau. The optimal sequence at $L = 12$, the

TABLE I. Optimal CHSH violation in the d_{1b} encoding (two generators AM, MB). $|S|$ is computed via the Horodecki bound from the maximal concurrence C .

L	$ S $	C	% Tsirelson	Sequence
3	2.199	0.457	77.7	ABB
4	2.225	0.488	78.7	ABAB
5	2.348	0.615	83.0	AABAB
6	2.398	0.662	84.8	ABBAAB
7	2.536	0.780	89.7	ABBAABB
8	2.540	0.783	89.8	BABBABAB
9	2.624	0.850	92.8	ABBAABABB
10	2.696	0.904	95.3	ABBAABBAAB
11	2.784	0.968	98.4	ABABBBBABAB
12	2.811	0.988	99.4	ABABABABABAB

alternating commutator structure $(\text{AM} \cdot \text{MB})^6$, achieves 99.4% of the Tsirelson bound.

We note that an earlier analysis [1] reported $|S| = 2.554$ and $C = 0.903$ at $L = 12$ for this encoding. That value was an optimization artifact: the original code employed six fixed angular starting points and a 9^4 grid for CHSH angle optimization, which missed the global maximum. Our Horodecki-based evaluation, which requires only the concurrence (a basis-independent quantity), recovers the true optimum.

B. Three-generator landscape

Table II presents the analogous results for the three-generator encoding, enumerated over 3^L sequences for $L = 2$ –8.

TABLE II. Optimal CHSH violation in the three-generator encoding $(\sigma_2, \sigma_3, \sigma_4)$ under 4D computational-subspace projection (Protocol A, Sec. V). The $|S|$ column is the exhaustive-enumeration maximum $S_{\max}^{4D}$ from the canonical `landscape_validation.json`; $C = \sqrt{(|S|/2)^2 - 1}$; % Tsirelson = $|S|/2\sqrt{2}$.

L	$ S $	C	% Tsirelson	Sequence
2	2.000	0.000	—	(no entanglement)
3	2.722	0.923	96.22	(enumeration max)
4	2.722	0.923	96.22	(enumeration max)
5	2.791	0.973	98.68	(enumeration max)
6	2.791	0.973	98.68	(enumeration max)
7	2.791	0.973	98.68	(enumeration max)
8	2.824	0.997	99.84	(enumeration max)

The landscape approaches Tsirelson saturation ($|S| = 2\sqrt{2}$, $C = 1$) already at $L = 8$, reaching $|S| = 2.824$ ($C = 0.997$, 99.84% of the Tsirelson bound) at the enumeration maximum; exhaustive search over all $3^8 = 6561$

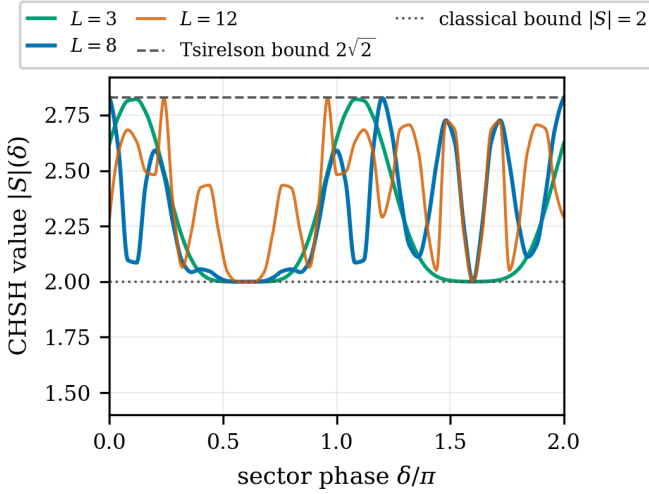


FIG. 1. Sector-phase dependence $|S|(\delta)$ of the optimal three-generator braiding sequence at braid lengths $L = 3, 8, 12$. The CHSH value swings between the classical bound $|S| = 2$ in the “dead” sector phases and the Tsirelson bound $2\sqrt{2}$ (dashed) where the entangling sequence is aligned with the sector basis; the landscape develops finer structure at larger L , consistent with the decreasing Fourier-fit quality (Sec. VII). Curves are shape-preserving (PCHIP) interpolations through the $N = 50$ computed sector-phase points; at $L = 12$ the sampling is coarse relative to the oscillation (cf. the low Fourier-fit R^2), so structure finer than the sampling is not resolved. Data: `s_delta_sweeps.json` (concept-DOI 10.5281/zenodo.19601352).

sequences finds no single braid word that saturates the bound exactly. The full sector-phase dependence $|S|(\delta)$ of these three-generator sequences is shown in Fig. 1.

At $L = 2$, no sequence produces entanglement ($C = 0$), because a single application of any one generator cannot cross the bipartition. From $L = 3$ onward, the concurrence exceeds 0.92 (rising to 0.997 at $L = 8$), reflecting the availability of σ_3 as a primitive entangling gate.

The CHSH-violation fraction under 4D projection rises monotonically: 44% at $L = 3$, 79% at $L = 6$, 92% at $L = 9$ (see Sec. V for the protocol dependence of these fractions).

C. Why two encodings converge

The two encodings differ not in the reachable group but in the *word length* required to approximate a given unitary. Both $\{\text{AM}, \text{MB}\}$ and $\{\sigma_2, \sigma_3, \sigma_4\}$ generate dense subgroups of $\text{SU}(4)$ acting on the computational subspace [5].

The three-generator set approaches the Tsirelson bound already at $L = 8$ because σ_3 directly crosses the Alice–Bob bipartition, providing the entangling gate in the standard *rotate–entangle–rotate* decomposition. The two-generator d_{1b} set must synthesize entanglement indirectly through the mediator, requiring approximately

1.5 times more braid steps to reach the same concurrence. At sufficient length, both encodings converge to the same maximal violation—the distinction is one of *circuit depth*, not reachable dynamics. The representation-dependence of the CHSH operator norm on higher-dimensional Hilbert spaces is discussed in Ref. [21].

IV. TOPOLOGICAL SELECTION: THE σ_3 SWITCH

We now demonstrate that a single topological degree of freedom controls the entire CHSH landscape.

A. Blocking the cross-bipartition generator

In the three-generator encoding, we replace σ_3 ¹ by the 5×5 identity matrix and evaluate the concurrence on three representative $\{\sigma_2, \sigma_4\}$ configurations, each swept over 50 linearly spaced sector-phase values $\delta \in [0, 2\pi]$ (150 evaluations in total).

The result is absolute:

$$C(\mathcal{B}, \delta)|_{\sigma_3 \rightarrow I} = 0 \quad \forall \mathcal{B}, \forall \delta. \quad (5)$$

Across all tested blocked configurations (three representative sequences at $L = 3, 6, 8$ plus the d_{1b} winner at $L = 12$ with each of its two generators blocked in turn, each configuration swept over 50 sector phases, totaling $5 \times 50 = 250$ evaluations), the maximal concurrence is $C < 4.4 \times 10^{-16}$ —machine zero. Figure 2 contrasts the two cases for a representative $L = 8$ sequence $\sigma_2\sigma_3\sigma_4^4\sigma_3\sigma_2$: with σ_3 active, $|S|(\delta)$ rises well above the classical bound across the sector phase, approaching the Tsirelson regime; with σ_3 blocked, it is pinned to the classical bound for every δ .

B. Physical mechanism

The vanishing of entanglement under σ_3 -blocking has a transparent algebraic origin: σ_2 preserves the x_3 label (Bob’s qubit) and σ_4 preserves the x_1 label (Alice’s qubit). Without σ_3 , which re-fuses the intermediate charge x_2 and thereby couples the two branches of the fusion tree, the generators act independently on the two logical qubits. The resulting state is necessarily a product state, regardless of sequence length or sector phase.

¹ Index convention across the companion series: with $n = 6$ anyons and the bipartition cut after anyon 3, the cross-bipartition generator of this paper is σ_3 , while σ_4 acts locally on Bob’s side. In the companion paper [2] ($n = 8$ anyons, cut after anyon 4) the cross-bipartition role is played by its σ_4 ; the index shift is purely notational.

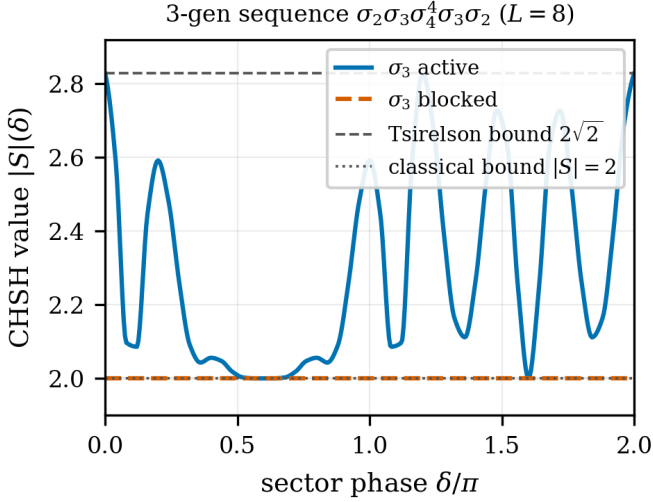


FIG. 2. The σ_3 generator as a topological switch, for a representative $L = 8$ three-generator sequence $\sigma_2\sigma_3\sigma_4^4\sigma_3\sigma_2$. With σ_3 active (solid), $|S|(\delta)$ rises well above the classical bound across the sector phase, approaching the Tsirelson regime; replacing σ_3 by the identity (dashed) makes the concurrence vanish identically [Eq. (5)], pinning $|S|$ to the classical bound $|S| = 2$ for every δ . Data: `topology_switch.results.json` (concept-DOI 10.5281/zenodo.19601352).

C. Encoding independence

The same test in the d_{1b} encoding yields identical results: blocking either AM or MB collapses C to zero exactly for the optimal sequence $(\text{AM} \cdot \text{MB})^6$ at all 50 sector phases.

This encoding independence confirms that the σ_3 -blocking result is a property of the *Fibonacci TQFT fusion rules*, not of a particular qubit encoding.

D. Interpretation

The σ_3 switch establishes that topology acts as a *gatekeeper* for entanglement: it does not create or destroy entanglement, but determines whether entanglement is *accessible* through the available braiding operations. This is the first pillar of the trio argument completed in Sec. VI.

In combination with the mutual-information results of Ref. [1], where topology was reported to rotate the effective measurement basis, the present finding demonstrates a stronger form of topological control: a single topological pathway decides yes-or-no whether CHSH violation can occur.

V. MEASUREMENT PROTOCOL AND COMPUTATIONAL-SUBSPACE LEAKAGE

The CHSH-violation fractions reported in Sec. III for the three-generator encoding depend on how observables are defined in the five-dimensional fusion space. We distinguish two measurement protocols and present results for both.

A. Protocol A: Computational-subspace projection

The four-dimensional computational subspace is defined by identifying the logical two-qubit basis as $|00\rangle \leftarrow |0\rangle$, $|01\rangle \leftarrow |1\rangle$, $|10\rangle \leftarrow |3\rangle$, and $|11\rangle \leftarrow (|2\rangle + |4\rangle)/\sqrt{2}$. The last mapping coherently recombines the two basis states with $x_1 = \tau$, $x_3 = \tau$ that differ only in the intermediate charge x_2 . After projection and renormalization, the standard two-qubit Horodecki criterion (4) applies.

Under Protocol A, the CHSH-violation fraction rises monotonically with braid length and reaches 91.8% at $L = 9$ (Table III). The maximal CHSH value under Protocol A saturates at $|S| = 2.824$ (Table II), within 0.16% of the Tsirelson bound $|S| = 2\sqrt{2}$ —approached but, per the exhaustive 3^8 -sequence search, not exactly attained by any single braid word at $L = 8$, reflecting the residual cost of the four-dimensional computational-subspace projection rather than a different physical limit.

B. Protocol B: Native 5D CHSH

We define Alice and Bob observables directly in the five-dimensional fusion space:

$$Z_A = \text{diag}(+1, +1, -1, -1, -1), \quad (6)$$

$$Z_B = \text{diag}(+1, -1, -1, +1, -1), \quad (7)$$

assigning eigenvalues ± 1 according to the x_1 and x_3 labels. The X -observables implement spin flips within each qubit sector: X_A swaps $|0\rangle \leftrightarrow |3\rangle$ and $|1\rangle \leftrightarrow |4\rangle$, while X_B swaps $|0\rangle \leftrightarrow |1\rangle$ and $|3\rangle \leftrightarrow |4\rangle$. The state $|2\rangle = (\tau, 1, \tau)$ has no flip partner in either qubit sector, giving $X|2\rangle = 0$ —i.e., the X and Y observables have spectrum $\{-1, 0, +1\}$ (trit-valued).

The CHSH maximum is computed via the Horodecki formula on the 3×3 correlation matrix $T_{ij} = \text{Re} \langle \psi | A_i \otimes B_j | \psi \rangle$, yielding $|S|_{\max} = 2\sqrt{m_1 + m_2}$ where m_1, m_2 are the two largest eigenvalues of $T^T T$.

All observables commute across the bipartition ($[A_i, B_j] = 0$ for all i, j), as required for a valid Bell test.

C. Protocol comparison

Table III compares the two protocols across braid lengths $L = 3-9$ (exhaustive enumeration of all 3^L se-

quences).

TABLE III. CHSH-violation statistics under two measurement protocols. $|S|_{\max}$ is the largest CHSH value across all sequences. CHSH% counts sequences with $|S| > 2$. $N_{\min}^2$ is the minimum computational-subspace fidelity.

L	$ S _{\max}^{5D}$	$ S _{\max}^{4D}$	CHSH% _{5D}	CHSH% _{4D}	Δ	$N_{\min}^2$
3	2.305	2.722	25.9	44.4	+18.5	0.736
4	2.305	2.722	59.3	61.7	+2.5	0.635
5	2.517	2.791	60.1	72.4	+12.3	0.480
6	2.517	2.791	60.9	79.0	+18.1	0.281
7	2.517	2.791	64.2	84.2	+20.0	0.281
8	2.517	2.824	64.9	88.5	+23.6	0.183
9	2.517	2.824	64.6	91.8	+27.1	0.131

Figure 3 summarizes both protocols across braid length. The two protocols agree qualitatively: under both, a majority of braiding sequences violate the CHSH inequality for $L \geq 4$. However, they differ quantitatively in two respects:

(i) The maximal CHSH value in Protocol B saturates at $|S|_{\max}^{5D} \approx 2.517$ from $L = 5$ onward, while Protocol A approaches the Tsirelson bound (within 0.16%). The gap reflects the zero eigenvalue of the X and Y observables on $|2\rangle$: a maximally entangled singlet-like state cannot be fully realized in the five-dimensional space using trit-valued observables.

(ii) The CHSH fraction in Protocol B plateaus near 64–65%, compared to the monotonically rising fraction under Protocol A.

D. Leakage statistics

The computational-subspace fidelity is the squared overlap onto the orthonormal logical basis $\{|0\rangle, |1\rangle, |3\rangle, (|2\rangle + |4\rangle)/\sqrt{2}\}$,

$$N^2 = |\psi_0|^2 + |\psi_1|^2 + |\psi_3|^2 + \frac{1}{2}|\psi_2 + \psi_4|^2, \quad (8)$$

which measures the fraction of the five-dimensional state that survives projection into the two-qubit subspace and satisfies $N^2 \in [0, 1]$ by construction. At $L = 9$, the minimum fidelity across all 19 683 sequences is $N_{\min}^2 = 0.131$, meaning that for some sequences up to 86.9% of the amplitude resides outside the computational subspace.

This is consistent with the expectation that exact leakage-free entangling gates may not exist for Fibonacci anyons [8]. Protocol A implicitly post-selects on the computational subspace by renormalization; Protocol B avoids this at the cost of a reduced maximal $|S|$.

E. Summary statement

CHSH violation is generic under *both* measurement protocols. The qualitative finding—that a major-

ity of Fibonacci braiding sequences violate the CHSH inequality—is protocol-independent. The quantitative CHSH fraction (64.6% vs. 91.8% at $L = 9$) and the saturation value of $|S|_{\max}$ (2.517 vs. 2.824) are protocol-dependent, reflecting the incomplete flip symmetry of the five-dimensional fusion space.

Note on terminology. In this paper, “Protocol B” refers specifically to the bipartite native-5D CHSH evaluation with trit-valued Z and X -like observables on the full five-dimensional fusion space (Sec. V, this paper). Other companion analyses that probe alternative non-bipartite CHSH constructions on the same fusion space (e.g., minimal three-diagonal-parity single-particle constructions) are distinct in scope and are not covered by the present landscape; the bipartite Protocol B used here remains well-defined as a CHSH evaluation in its own right.

We emphasize that the d_{1b} encoding (Sec. III, Table I) operates natively in four dimensions with no leakage and yields the headline $|S| = 2.811$ figure of the present landscape, the same value adopted as the Fibonacci reference in Ref. [1].

VI. SEQUENTIAL BRAIDING AND TEMPORAL CORRELATIONS

We now complete the trio argument by demonstrating that sequential braiding on a shared fusion space produces measurable temporal correlations in the CHSH-value series.

A. Setup

A register of six Fibonacci anyons is initialized in the vacuum state $|0\rangle$ and subjected to $N = 1000$ sequential braiding steps. At each step n , a generator $g_n \in \{\sigma_2, \sigma_3, \sigma_4\}$ is drawn from a specified distribution, the state is updated as $|\psi_n\rangle = g_n|\psi_{n-1}\rangle$, and its native five-dimensional (Protocol B, Sec. V) CHSH value $S_n = S[\psi_n]$ is recorded. This is repeated for $M = 20$ independent runs.

Three models are compared:

Model A (i.i.d.): each generator is chosen uniformly at random ($p = 1/3$ each).

Model B (Markov-persistent): with probability 0.6, the previous generator is repeated; otherwise a uniform draw from the remaining two.

Model C (no σ_3): generators drawn from $\{\sigma_2, \sigma_4\}$ only ($p = 1/2$ each), serving as a control.

B. Pillar 2: Null test

As a prerequisite, we verify that *independent* sector phases produce no autocorrelation. Drawing $N = 10\,000$

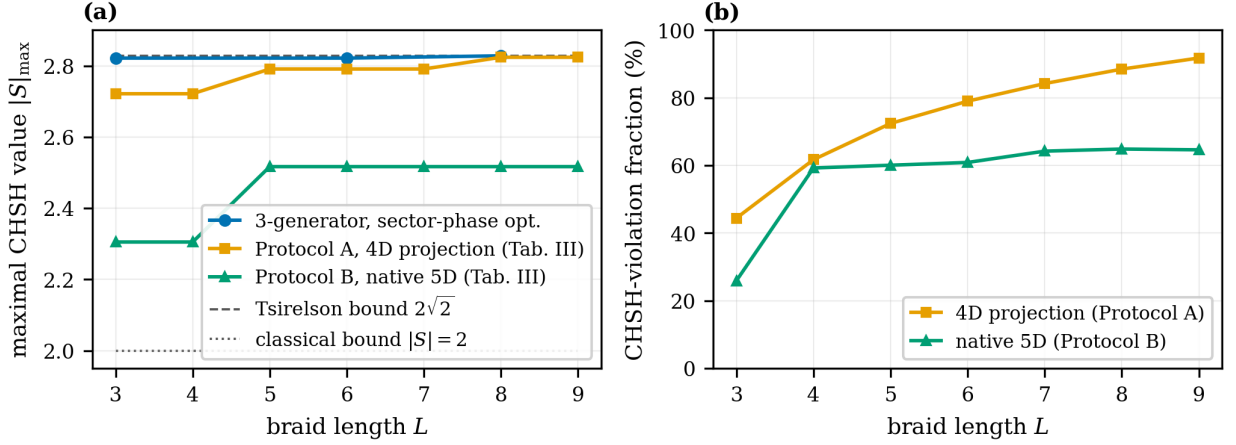


FIG. 3. Braiding-length dependence of the CHSH landscape. (a) Maximal CHSH value $|S|_{\max}$ versus braid length L for the three-generator encoding at the sector-phase optimum of the deposited winning sequences (per- L maxima of `s_delta_sweeps.json`, approaching the Tsirelson bound at $L = 8$), and for the two operational measurement protocols (Table III): Protocol A (four-dimensional computational-subspace projection, approaching $2\sqrt{2}$ within 0.16%) and Protocol B (native five-dimensional, trit-valued observables, saturating near 2.517). Dashed: Tsirelson bound $2\sqrt{2}$; dotted: classical bound $|S| = 2$. (b) Fraction of braiding sequences with $|S| > 2$ under the two protocols, rising to 91.8% (Protocol A) and plateauing near 64.6% (Protocol B) at $L = 9$. Data: `landscape_validation.json` and `s_delta_sweeps.json` (concept-DOI 10.5281/zenodo.19601352).

sector phases δ_i uniformly from $[0, 2\pi]$ and computing the CHSH value for each, the lag-1 autocorrelation is $\text{ACF}(1) = +0.001$ (Model A configuration) and $\text{ACF}(1) = +0.010$ (d1b configuration), both within the 95% confidence interval $\pm 1.96/\sqrt{N} = \pm 0.020$ (seeded evaluation deposited as `d8_acf_null_robustness.py`).

The null hypothesis ($\text{ACF} = 0$ under i.i.d. sectors) is not rejected. This confirms that any nonzero autocorrelation observed in the sequential-braiding models is a genuine dynamical effect, not a statistical artifact.

C. Pillar 3: Sequential-braiding autocorrelation

Table IV presents the measured autocorrelation for the three models.

TABLE IV. Lag-1 autocorrelation of the CHSH time series under sequential braiding. Significance is measured against the shuffle confidence band (1000 random permutations of the observed series).

Model	Generators	ACF(1)	Significance	CHSH%
A (i.i.d.)	$\sigma_2, \sigma_3, \sigma_4$	+0.343	10.85σ	99.2
B (Markov)	$\sigma_2, \sigma_3, \sigma_4$	+0.361	11.24σ	98.6
C (no σ_3)	σ_2, σ_4	0.000	0.00σ	0.0

Both Models A and B exhibit highly significant positive autocorrelation, with Model B only marginally above Model A (0.361 vs. 0.343). This indicates that the autocorrelation arises primarily from *state continuity*—each braid step applies a finite rotation, keeping $|\psi_n\rangle$ close

to $|\psi_{n-1}\rangle$ in Hilbert space—rather than from generator-level persistence.

Model C produces identically zero autocorrelation, confirming the σ_3 -blocking result of Sec. IV in a time-resolved setting: without the cross-bipartition generator, the CHSH series is constant at $|S| = 2.000$ with zero variance.

The runs test reinforces the clustering structure: in Model A, only 1 crossing of the $|S| = 2$ threshold is observed per run, compared to an i.i.d. expectation of approximately 16.

D. Robustness

Replacing the vacuum initial state with a seeded random normalized five-dimensional vector yields $\text{ACF}(1) = +0.318$ (Model A) and $+0.326$ (Model B), confirming that the effect survives initialization.

The full autocorrelation range across the four deposited setups (vacuum and random initialization, Models A and B) is $\text{ACF}(1) \in [0.31, 0.37]$, with the representative value for long runs (Model A, vacuum start) being $+0.34$.

E. The trio argument

The three pillars combine to establish that the topological sector structure is real, measurable, and not attributable to chance:

Pillar 1 (Sec. IV): σ_3 -blocking shows that a single generator switches entanglement on and off—topology con-

trols access.

Pillar 2 (this section): the null test confirms that i.i.d. sector phases produce no autocorrelation—the method has no intrinsic bias.

Pillar 3 (this section): sequential braiding produces $\text{ACF}(1) = +0.34$ at $> 10\sigma$ —the sector structure generates genuine temporal correlations without additional assumptions.

F. Falsifiable prediction

In an experiment with sequential anyon braiding, the detection sequence of CHSH-violation outcomes should exhibit positive autocorrelation. We define the signed per-run CHSH term

$$g_n = (-1)^{a_n b_n} x_n y_n \in \{-1, +1\}, \quad (9)$$

where $a_n, b_n \in \{0, 1\}$ label the measurement settings and $x_n, y_n \in \{-1, +1\}$ the outcomes of run n ; $\text{ACF}(1)$ denotes the lag-1 autocorrelation of the g_n series. Standard quantum mechanics—under the assumption of independent measurement choices between runs—predicts no systematic positive ACF beyond statistical fluctuations. The specific value ($\text{ACF}(1) \approx 0.34$) is model-dependent, but the qualitative prediction $\text{ACF} > 0$ is a signature specific to *sequential* braiding on a shared fusion register, distinct from the spacelike-separated, independently re-prepared trials of loophole-free Bell-test data such as those of Refs. [12, 13, 33].

As a null control, we computed $\text{ACF}(1)$ on the published outcome sequence of the openly available loophole-free test of Hensen *et al.* [12] (data DOI 10.4121/uuid:6e19e9b2-4a2d-40b5-8dd3-a660bf3c0a31, $N = 245$ valid trials), finding $\text{ACF}(1) = -0.047$ (-0.64σ , $|\text{ACF}(1)| < 0.125$ at 2σ)—consistent with zero, as expected: the sequential-braiding mechanism proposed here is structurally absent from independently re-prepared Bell trials, so this is a genuine null control rather than a failed prediction. The earlier test of Giustina *et al.* [13] is closed as a venue for this analysis, as its outcome sequence is available only on request; the higher-statistics Shalm–NIST test [33], although also openly available, was not analyzed here. The appropriate venue to test the sequential-braiding prediction is a platform that physically implements braiding sequentially on a shared register, such as the digital Fibonacci-anyon realization of Xu *et al.* [43] or the digital non-Abelian Ising-anyon braiding of Andersen *et al.* [11]; both are gate-based (digital) realizations rather than the analog continuous-braiding transport assumed in the present model, a distinction relevant to any quantitative comparison. The memory loophole in sequential Bell-type data is well characterized [34]; related but distinct analyses of the same loophole-free data sets include a no-signaling test [35] and a Kolmogorov-complexity analysis of the outcome sequences [36, 37], neither of which computes the lag-1 ACF proposed here; improved

statistical methodology for Bell-experiment data is discussed in Ref. [44].

VII. DISCUSSION

A. CHSH violation is generic

The central finding of this work is that CHSH violation is not a fine-tuned property of special braiding sequences but a generic feature of the Fibonacci anyon landscape. Under both measurement protocols, more than half of all braiding sequences violate the CHSH inequality for braid lengths $L \geq 4$. The optimal sequences exhibit palindromic structure in both encodings, and the three-generator necessity (with two generators: $C = 0$ always) provides a clean algebraic explanation for why entanglement requires topological connectivity across the bipartition.

B. Topology as gatekeeper

The σ_3 -blocking result and the sequential-braiding autocorrelation together establish a picture in which topology does not create or destroy entanglement, but *gates* it: the topological pathway governs whether entanglement is accessible through the available braiding operations. This complements the mutual-information study of Ref. [1], where topological sector labels were found to correlate with Bell-type observables in finite lattice models (there reported as a finite-size effect).

C. Implications for topological quantum computation

The σ_3 -blocking mechanism has a direct interpretation in the context of topological quantum computing: any braiding protocol that inadvertently excludes the cross-bipartition generator will produce zero entanglement, regardless of circuit depth. The leakage analysis (Sec. V) shows that this concern extends beyond gate selection to the measurement protocol, as up to 86.9% of state weight can reside outside the computational subspace. This reinforces the finding of Cui *et al.* [8] that exact leakage-free entangling gates for Fibonacci anyons may not exist.

D. Limitations

Several limitations qualify our results:

(i) All computations are *numerical simulations* within the idealized Fibonacci TQFT at zero temperature and zero decoherence. The charge-parity correlator underlying the CHSH violation is topologically robust against braiding-phase noise (dephasing) [3]: for a closed-form two-parameter degradation model $|S| = 2\lambda^2\sqrt{1+v^4C^2}$

(charge-survival probability λ , dephasing visibility v , ideal concurrence C), pure dephasing ($v < 1$, $\lambda = 1$) bounds $|S|$ from below by the classical value 2 (attained only in the fully-dephased limit $v \rightarrow 0$) and never drives it below—the violation *vanishes* rather than surviving degraded. Only charge-changing quasiparticle poisoning ($\lambda < 1$) can push $|S|$ below 2, at the Werner-state threshold [7] $\lambda^* = 2^{-1/4} = 0.8409$ for an ideal ($C = 1$) Bell state and $\lambda^* = 0.8416$ for the actually braided state ($C = 0.9968$)—a poisoning tolerance of order 16% per qubit before the classical bound is restored; this follows directly from the Horodecki and Werner criteria and is not a new bound. A gap-scaling trade-off follows: the thermal quasiparticle poisoning rate is exponentially suppressed by the topological gap [27] (Arrhenius factor $e^{-\Delta/k_B T}$), so the universal $\nu = 12/5$ Fibonacci candidate—with a measured activation gap of order tens of mK [29], the smallest of the accessible non-Abelian platforms, while the $\nu = 5/2$ Ising gap is comparatively robust but not computationally universal. A nonequilibrium quasiparticle floor can exceed the equilibrium poisoning rate by orders of magnitude in realistic samples and is decoupled from temperature, so the limitation is not purely thermal and does not vanish as $T \rightarrow 0$. The idealized $k = 3$ Read–Rezayi identification of $\nu = 12/5$ is itself optimistic in one respect (particle-hole conjugation and Landau-level mixing strengthen the effective gap [28]) and pessimistic in another (companion anyons at nearby fillings, observed in the first excited Landau level [32], add poisoning channels). We do not quote a specific operation-time bound: non-Abelian braiding at $\nu = 12/5$ has not been experimentally demonstrated, so any timescale estimate would be conditional on an identification— $\nu = 12/5$ as the Fibonacci-anyon $k = 3$ Read–Rezayi state—that is numerically supported but not experimentally confirmed [28, 30, 31]. Measurement imperfections beyond this poisoning model are not addressed here.

(ii) The sector phase δ entering via $R_\tau \rightarrow R_\tau \cdot e^{i\delta}$ parameterizes a family of braid representations but is not derived from a first-principles Chern-Simons path integral.

(iii) The CHSH-violation fraction is protocol-dependent (Sec. V): 64.6% and 91.8% represent different—both legitimate—definitions of measurement on the fusion space.

(iv) The autocorrelation prediction (Sec. VI) is quantitatively setup-dependent ($\text{ACF}(1) \in [0.31, 0.37]$ across the four deposited setups), though qualitatively robust ($\text{ACF} > 0$).

(v) The $S(\delta)$ functional form becomes increasingly complex at longer braid lengths: a six-term Fourier fit achieves $R^2 = 0.997$ at $L = 3$ but only $R^2 = 0.38$ at $L = 12$ in the five-dimensional encoding, indicating that the CHSH landscape develops structure beyond simple harmonic description.

E. Comparison with the independent fusion-sector construction of Xu, Ye, and You

Independently and contemporaneously, Xu, Ye, and You [20] have studied CHSH correlations on triplets of the Fibonacci-anyon state $|\tau, \tau; 1\rangle$ using a graded three-dimensional fusion-sector Hilbert space $\mathcal{H}_1 \oplus \mathcal{H}_\tau$ with measurement operators of the form $A_\alpha = \text{diag}(1, [\cos \alpha, \sin \alpha], [\sin \alpha, -\cos \alpha])$ (their Eq. (9)). Their construction reaches $|S|_{\max} = (4\sqrt{2}\varphi + 2)/\varphi^3 \approx 2.633$ with superactivation as the underlying phenomenon: a single copy of $|\tau, \tau; 1\rangle$ admits a local hidden-variable model, while three copies activate Bell nonlocality. Their setup is bipartite under the graded tensor product, with A_α block-diagonal across $\mathcal{H}_1$ and $\mathcal{H}_\tau$. Tsirelson’s bound $2\sqrt{2}$ applies in their setting; the value 2.633 is sub-Tsirelson due to nonuniform Schmidt weights of the three-copy state (weight $2/\varphi^2 \approx 0.764$ on $\mathcal{H}_\tau \otimes \mathcal{H}_\tau$).

The present paper’s d_{1b} four-dimensional encoding, by contrast, operates on a single copy of an entangled four-dimensional fusion state with concurrence $C = 0.988$, reaching $|S| = 2.811$ at $\delta/\pi = 1.76$, which saturates the bipartite Horodecki bound $2\sqrt{1+C^2}$ for this concurrence. The two constructions address different questions (multi-copy activation versus single-copy braid-induced violation) and produce different but compatible values: both respect Tsirelson’s bound, both bipartite Bell-operator norms equal $2\sqrt{2}$ exactly, and the difference reflects distinct Schmidt structures of the underlying states. An independent reproduction of both values, together with the $|S| = 3.406$ value from the companion paper [2] (which uses a nonbipartite construction violating setting-locality, not subject to Tsirelson’s bound), is documented in a dedicated comparison analysis of the project repository: all three values are reproduced to within their natural precisions, Bell-operator-norm comparisons confirm the structural distinction, and the standard sanity tests (state norm, outcome-probability summation, operator-norm vs algebraic bound) exclude post-selection or nonstandard normalization in any of the three constructions.

F. Outlook

The present enumeration is bounded at $L = 12$ by the 3^L scaling; other anyon models (Ising, $\text{SU}(2)_k$ for $k > 3$) and experimental autocorrelation data are not addressed here. Whether the Kitaev honeycomb model [14], which offers a lattice setting in which both classical and quantum limits are accessible, can bridge the present work with the $\mathbb{Z}_2$ results of Ref. [1] remains an open question. A further open question, suggested by the Schmidt-rank-3 structure of the Xu, Ye, You construction [20], is whether a single-copy Schmidt-rank-3 state in the four-dimensional d_{1b} Hilbert space (or in the five-dimensional fusion space) could be prepared by a different braiding protocol; whether such a state would yield CHSH values

intermediate between the present results remains open.

VIII. CONCLUSION

We have presented the first, to our knowledge, complete sequence-by-sequence enumeration of CHSH violation across all Fibonacci anyon braiding words of length $L = 3$ –12 in two complementary encodings, in the broader landscape of qualitative Bell-test constructions on Fibonacci/Ising and Majorana platforms [15–19] and the independently and contemporaneously analyzed three-copy fusion-sector construction of Xu, Ye, and You [20]; a companion Leggett–Garg-inequality letter on the same Fibonacci representation, saturating the Lüders bound $K_3 \rightarrow 3/2$, is reported in Ref. [22].

CHSH violation is generic: 64.6% of sequences violate the CHSH inequality under native five-dimensional measurement, and 91.8% under computational-subspace projection, at $L = 9$.

The optimal d_{1b} sequence at $L = 12$ reaches $|S| = 2.811$ (99.4% of the Tsirelson bound $2\sqrt{2}$, $C = 0.988$), while the three-generator encoding on the full five-dimensional fusion space approaches the Tsirelson bound ($|S| = 2.824$, 99.84%) already at $L = 8$.

Topology acts as a gatekeeper through the σ_3 switch: removing the cross-bipartition generator collapses entanglement to zero exactly, independent of sequence length, sector phase, and encoding.

Sequential braiding on a shared anyon register produces positive lag-1 autocorrelation ($\text{ACF}(1) = +0.34$ at 10.85σ), a falsifiable signature of state continuity that, under the assumption of independent measurement settings between runs, is not predicted by standard quantum mechanics.

We emphasize that the CHSH values reported here, while in excess of the classical bound, should not be interpreted as Bell violations in the EPR sense (cf. Sec. II E): the underlying Fibonacci fusion space does not factorize as a tensor product of local Alice and Bob subsystems, and the classical-bound excess is more precisely characterized as a signature of topological nonseparability in fusion space.

Together with the finite-size mutual-information correlations reported in Ref. [1], these findings indicate that the available topological operations gate whether CHSH correlations are accessible—a structural pattern seen, with the caveats noted there, in finite classical lattice systems and in quantum anyon systems alike.

DATA AND CODE AVAILABILITY

The deterministic scripts and JSON output files that validate the results reported in this paper are archived on Zenodo under the concept DOI [10.5281/zenodo.19601352](https://doi.org/10.5281/zenodo.19601352) (which always resolves to the latest version) and released under the Creative Commons Attribution 4.0

International (CC BY 4.0) license. The archive includes the three-generator (5D) landscape validation and its leakage-norm reanchor (`landscape_validation.py`, `n2_reanchor.py`); a closed-form sector-phase sweep validation covering all rows of Table I and the sequences of Table II (`d4_s_delta_sweep_validation.py`, reproducing `s_delta_sweeps.json` and the printed table values); the topological-gatekeeper validation, including the exact σ_3 -block $C = 0$ check (`d5_topology_switch_validation.py`, reproducing `topology_switch_results.json`); the sequential-braiding autocorrelation prediction of Table IV (`d6_acf_prediction_sequential_braiding.py`); the seeded Pillar-2 null test and initialization-robustness evaluation of Sec. VI (`d8_acf_null_robustness.py`); and an independent null-control reanalysis of the Hensen *et al.* (2015) loophole-free Bell data, including the underlying public dataset (`d7_acf_hensen2015_null_control.py`). The exhaustive sequence search that originally identified the winning braid words reported in Tables I and II, and the verification scripts for the sector-phase Yang–Baxter check of Appendix C (Table V), are not part of this record.

USE OF AI TOOLS

In the research, processing, and writing of this paper and its results I worked together with generative AI tools, in practice a system of multiple coordinated AI instances that I set up and orchestrate (large language models, mainly Claude, by Anthropic, inside Claude Code). At their current context sizes I found it far more effective to work with several specialized instances, each with its own role and its own harness of rules and parameters that I designed and refined through feedback, than to load a single instance with all of the material; for my workflow that would have been inefficient, though this depends on the individual implementation. I lead this collaboration: I choose the research directions, set the goals, and make the final decisions in open exchange with the AI, learning actively as the work proceeds. The AI carries out the drafting, including the mathematical and technical parts, the numerical computation, and the literature search, under my direction. The AI works autonomously only task by task, within the structure I develop through feedback: it completes a task, and at open questions that need me it stops until the point is settled before the next step. Along the way I witness and take many of the decisions that shape the path, and it is common for me to spot things that need improvement. The work spans many separate runs, and a single simulation or build task alone can take up to an hour, so it could not happen all together in one autonomous run; and had I let the AI do all of it together alone, even if it is possible, it would no longer be my work but the AI’s.

I run multiple verifications at the different stages of the

work and one before release, including cross-checks with an unrelated AI model from a different company, and all references are checked against the original sources. In the end what matters are human eyes, a principle that is itself written into the parameters of my system: I reach out to experts after publishing for review and feedback, so I learn what is solid and what must be corrected or falsified. My scripts for reproduction and review are released with this record. These tools are not authors; I am the author, and I take full responsibility for all scientific content and decisions leading to these results and their publication.

ACKNOWLEDGMENTS

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. The author declares no competing interests.

Appendix A: Braid-generator matrices

The five braid generators $\sigma_1, \dots, \sigma_5$ for six Fibonacci anyons in the basis (1) are constructed as follows. For each generator σ_k , the anyon pair (τ_k, τ_{k+1}) is braided via the F -move/ R -phase/ F -move sequence on the relevant two-dimensional subspace of the fusion tree. The remaining basis states acquire diagonal R -phases. Explicit 5×5 matrices are provided in the supplementary code archived with the Zenodo deposit.

The key structural properties:

- σ_2 mixes states within the $x_1 \in \{1, \tau\}$ subspace (Alice’s qubit), preserving x_3 .
- σ_4 mixes states within the $x_3 \in \{1, \tau\}$ subspace (Bob’s qubit), preserving x_1 .
- σ_3 re-fuses the intermediate label x_2 , coupling both branches of the fusion tree.
- σ_1 and σ_5 act as diagonal phases on x_1 and x_3 respectively.

Appendix B: Fourier structure of $S(\delta)$

The CHSH value as a function of sector phase δ exhibits encoding-dependent harmonic structure. For the Ising anyon model, the functional form is exactly $S(\delta) = 2\sqrt{2}\cos^2(\delta/2)$, containing only the fundamental harmonic $n = 1$. This idealized Ising result assumes access to a nonstabilizer (“magic”) single-qubit operation beyond the topologically protected Clifford gate set; purely stabilizer operations on Ising anyons cannot violate the CHSH inequality [42]. For Fibonacci anyons, the

leading nonconstant harmonic of a six-term Fourier fit at $L = 3$ ($\sigma_4\sigma_2\sigma_3$ sequence, $R^2 = 0.997$) is $n = 4$:

$$\begin{aligned} S(\delta) \approx & 1.931 - 0.040 \cos \delta + 0.126 \sin \delta \\ & + 0.293 \cos 2\delta + 0.212 \sin 2\delta \\ & - 0.098 \cos 3\delta + 0.072 \sin 3\delta \\ & + 0.141 \cos 4\delta + 0.430 \sin 4\delta \\ & + 0.025 \cos 5\delta. \end{aligned} \quad (B1)$$

The apparent spectral contrast between Fibonacci (leading harmonic $n = 4$ in this fit) and Ising (exactly $n = 1$) is suggestive of order-dependent CHSH signatures, but the dominant-harmonic assignment is sensitive to the fit method and the sector-phase sampling (cf. the low fit quality at larger L below) and we do not pin it as a sharp invariant.

At longer braid lengths, R^2 decreases ($R^2 = 0.38$ at $L = 12$ in the 5D encoding), indicating that the $S(\delta)$ landscape develops structure beyond six-term Fourier description.

Appendix C: Verification of the five-dimensional representation under sector-phase modulation

The qualitative nonseparability and contextuality framing of Sec. II E and Sec. II D is supported by an independent numerical check of the braid-group relations for the full five-state generators $\sigma_1, \dots, \sigma_5$ across a range of sector phases δ . At $\delta = 0$ the standard Fibonacci modular tensor category is satisfied to floating-point precision (Yang–Baxter residuum $\leq 8.2 \times 10^{-16}$, distant commutators identically zero). Under the sector-phase modulation $R_\tau \mapsto R_\tau \cdot e^{i\delta}$ introduced in Sec. II D to parameterize the family of representations on the torus, Yang–Baxter deforms uniformly across all four neighboring generator pairs while distant commutators $[\sigma_i, \sigma_j] = 0$ for $|i-j| \geq 2$ remain identically zero throughout. Table V summarizes the verification. The sector phase δ is thus a free deformation knob rather than a physical parameter: generic $\delta \neq 0$ breaks the Yang–Baxter/hexagon consistency of the representation (Table V), and only $\delta = 0$ recovers the standard Fibonacci modular tensor category. This expected discreteness — rather than a continuum of consistent deformations — is consistent with the general Ocneanu-rigidity result for fusion categories [26], which here rests on the paper-internal numerics of Table V rather than on an independent categorical classification of this specific representation.

The intact $\delta = 0$ representation confirms that the failure of tensor-product factorization underlying the d_{1b} encoding (Sec. II D, Sec. II E) is a property of the deliberate four-dimensional projection of the five-dimensional fusion space, not of the underlying braid representation itself. The exact preservation of distant commutativity under δ -modulation is structural: the modulation acts locally on the R_τ eigenvalue and does not couple block- or diagonally-separated generators. The

TABLE V. Algebraic verification of the five-state representation under sector-phase modulation. The Yang–Baxter residuum $\|\sigma_i\sigma_{i+1}\sigma_i - \sigma_{i+1}\sigma_i\sigma_{i+1}\|_F$ is identical across all four neighboring pairs and is shown in column 2. The distant commutator $\|[\sigma_i, \sigma_j]\|_F$ for $|i - j| \geq 2$ is zero to all available numerical precision (exact, not float-floor) for every tested δ .

δ/π	YB residuum (5D)	distant commutator (5D)
0.000	$\leq 8.2 \times 10^{-16}$	0 (exact)
0.100	0.525	0 (exact)
0.250	0.958	0 (exact)
0.500	0.473	0 (exact)
1.000	0.873	0 (exact)
1.191	0.052	0 (exact)
1.500	1.523	0 (exact)

companion Leggett–Garg analysis [22] reports an analogous fixed-word vs envelope distinction for the temporal $K_3(\delta)$ periodicity. All values above were verified with the deterministic scripts `d2.delta.phase.consistency.py` and `d3.5d.braid.algebra.consistency.py`, which document the full underlying data including the $\mathbb{Z}_5$ phase spectrum of Δ^2 from the Birman formula and the d_{1b} spectator-filter algebra-breaks; these two scripts are not part of this record.

- [1] B. Y. Sayim, *Sector-Dependent CHSH Violation in Fibonacci Anyons, and Finite-Size Topological–CHSH Mutual Information in 2D Lattice Models*, Zenodo preprint (2026), [doi:10.5281/zenodo.19600752](https://doi.org/10.5281/zenodo.19600752).
- [2] B. Y. Sayim, *CHSH-Form Values Without Bell Non-locality: Inapplicability of the Tsirelson Bound on a Non-Factorized Fibonacci Fusion Space*, Zenodo preprint (2026), [doi:10.5281/zenodo.19601998](https://doi.org/10.5281/zenodo.19601998).
- [3] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, *Non-Abelian anyons and topological quantum computation*, Rev. Mod. Phys. **80**, 1083 (2008).
- [4] M. H. Freedman, A. Kitaev, M. J. Larsen, and Z. Wang, *Topological quantum computation*, Bull. Am. Math. Soc. **40**, 31 (2003).
- [5] N. E. Bonesteel, L. Hormozi, G. Zikos, and S. H. Simon, *Braid topologies for quantum computation*, Phys. Rev. Lett. **95**, 140503 (2005).
- [6] L. Hormozi, G. Zikos, N. E. Bonesteel, and S. H. Simon, *Topological quantum compiling*, Phys. Rev. B **75**, 165310 (2007).
- [7] R. Horodecki, P. Horodecki, and M. Horodecki, *Violating Bell inequality by mixed spin- $\frac{1}{2}$ states: necessary and sufficient condition*, Phys. Lett. A **200**, 340 (1995).
- [8] S. X. Cui, K. T. Tian, J. F. Vasquez, Z. Wang, and H. M. Wong, *The search for leakage-free entangling Fibonacci braiding gates*, J. Phys. A: Math. Theor. **52**, 455301 (2019); [arXiv:1904.01731](https://arxiv.org/abs/1904.01731).
- [9] P. C. Burke, C. Aravanis, J. Aspman, J. Mareček, and J. Vala, *Topological quantum compilation of two-qubit gates*, [arXiv:2408.07132](https://arxiv.org/abs/2408.07132) (2024).
- [10] Z.-C. Yang, K. Meichanetzidis, S. Kourtis, and C. Chamon, *Scrambling via braiding of nonabelions*, Phys. Rev. B **99**, 045132 (2019).
- [11] T. I. Andersen *et al.* (Google Quantum AI), *Non-Abelian braiding of graph vertices in a superconducting processor*, Nature **618**, 264 (2023); [arXiv:2210.10255](https://arxiv.org/abs/2210.10255).
- [12] B. Hensen *et al.*, *Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres*, Nature **526**, 682 (2015).
- [13] M. Giustina *et al.*, *Significant-loophole-free test of Bell’s theorem with entangled photons*, Phys. Rev. Lett. **115**, 250401 (2015).
- [14] A. Kitaev, *Anyons in an exactly solved model and beyond*, Ann. Phys. **321**, 2 (2006).
- [15] G. K. Brennen, S. Iblisdir, J. K. Pachos, and J. K. Slingerland, *Non-locality of non-Abelian anyons*, New J. Phys. **11**, 103023 (2009); [arXiv:0810.4319](https://arxiv.org/abs/0810.4319).
- [16] P.-Y. Chang, S.-K. Chu, and C.-T. Ma, *Bell’s inequality and entanglement in qubits*, J. High Energy Phys. **09** (2017) 100; [arXiv:1705.06444](https://arxiv.org/abs/1705.06444).
- [17] D. E. Drummond, A. A. Kovalev, C.-Y. Hou, K. Shtengel, and L. P. Pryadko, *Demonstrating entanglement by testing Bell’s theorem in Majorana wires*, Phys. Rev. B **90**, 115404 (2014); [arXiv:1403.0916](https://arxiv.org/abs/1403.0916).
- [18] A. Romito and Y. Gefen, *Ubiquitous nonlocal entanglement with Majorana zero modes*, Phys. Rev. Lett. **119**, 157702 (2017); [arXiv:1703.01841](https://arxiv.org/abs/1703.01841).
- [19] D. J. Clarke, J. D. Sau, and S. Das Sarma, *A practical phase gate for producing Bell violations in Majorana wires*, Phys. Rev. X **6**, 021005 (2016); [arXiv:1510.00007](https://arxiv.org/abs/1510.00007).
- [20] C.-Q. Xu, W. Ye, and L. You, *Superactivation of Bell nonlocality in pure anyonic states*, Phys. Rev. A **112**, 062232 (2025); [arXiv:2506.08919](https://arxiv.org/abs/2506.08919).
- [21] S. P. Sorella, *On the representations of Bell’s operators in Quantum Mechanics*, Found. Phys. **53**, 59 (2023); [arXiv:2304.05696](https://arxiv.org/abs/2304.05696).
- [22] B. Y. Sayim, *Leggett–Garg saturation and structural signatures in Fibonacci-anyon braiding*, Zenodo preprint (2026), [doi:10.5281/zenodo.20372744](https://doi.org/10.5281/zenodo.20372744).
- [23] M. S. Guimaraes, I. Roditi, and S. P. Sorella, *Bell-CHSH inequality and unitary operators*, Nucl. Phys. B (2024); [arXiv:2403.15276](https://arxiv.org/abs/2403.15276).
- [24] T.-M. Chou, *Topological Contextuality and Quantum Representations*, [arXiv:2506.14537](https://arxiv.org/abs/2506.14537) (2025).
- [25] P. Bonderson, C. Knapp, and K. Patel, *Anyonic Entanglement and Topological Entanglement Entropy*, [arXiv:1706.09420](https://arxiv.org/abs/1706.09420) (2017).
- [26] P. Etingof, D. Nikshych, and V. Ostrik, *On fusion categories*, [arXiv:math/0203060](https://arxiv.org/abs/math/0203060) (2002).
- [27] T. Karzig, W. S. Cole, and D. I. Pikulin, *Quasiparticle Poisoning of Majorana Qubits*, Phys. Rev. Lett. **126**, 057702 (2021).
- [28] R. S. K. Mong, M. P. Zaletel, F. Pollmann, and Z. Papić, *Fibonacci anyons and charge density order in the 12/5*

- and $13/5$ plateaus, Phys. Rev. B **95**, 115136 (2017); [arXiv:1505.02843](#).
- [29] C. Zhang, C. Huan, J. S. Xia, N. S. Sullivan, W. Pan, K. W. Baldwin, K. W. West, L. N. Pfeiffer, and D. C. Tsui, *Spin Polarization of the $12/5$ Fractional Quantum Hall Effect*, [arXiv:1204.0560](#) (2012).
- [30] K. Pakrouski, M. Troyer, Y.-L. Wu, S. Das Sarma, and M. R. Peterson, *The enigmatic $12/5$ fractional quantum Hall effect*, Phys. Rev. B **94**, 075108 (2016); [arXiv:1604.04610](#).
- [31] W. Zhu, S. S. Gong, F. D. M. Haldane, and D. N. Sheng, *The Fractional Quantum Hall States at $\nu = 13/5$ and $12/5$ and their Non-Abelian Nature*, [arXiv:1505.03050](#) (2015).
- [32] W. Pan, J. S. Xia, H. L. Stormer, D. C. Tsui, C. Vicente, E. D. Adams, N. S. Sullivan, L. N. Pfeiffer, K. W. Baldwin, and K. W. West, *Experimental studies of the fractional quantum Hall effect in the first excited Landau level*, Phys. Rev. B **77**, 075307 (2008); [arXiv:0801.1318](#).
- [33] L. K. Shalm *et al.*, *A strong loophole-free test of local realism*, Phys. Rev. Lett. **115**, 250402 (2015); [arXiv:1511.03189](#).
- [34] J. Barrett, D. Collins, L. Hardy, A. Kent, and S. Popescu, *Quantum nonlocality, Bell inequalities and the memory loophole*, Phys. Rev. A **66**, 042111 (2002); [arXiv:quant-ph/0205016](#).
- [35] G. Adenier and A. Yu. Khrennikov, *Test of the no-signaling principle in the Hensen “loophole-free CHSH experiment”*, Fortschr. Phys. **65**, 1600096 (2017); [arXiv:1606.00784](#).
- [36] M. G. Kovalsky, A. A. Hnilo, and M. B. Agüero, *Kolmogorov complexity of sequences of random numbers generated in Bell’s experiments*, Phys. Rev. A **98**, 042131 (2018); [arXiv:1805.07161](#).
- [37] M. G. Kovalsky, A. A. Hnilo, and M. B. Agüero, *Addendum to: Kolmogorov complexity of sequences of random numbers generated in Bell’s experiments (series of outcomes)*, [arXiv:1812.05926](#) (2018).
- [38] P. De Fabritiis, F. M. Guedes, M. S. Guimaraes, G. Peruzzo, I. Roditi, and S. P. Sorella, *Weyl operators, Tomita–Takesaki theory, and Bell–Clauser–Horne–Shimony–Holt inequality violations*, Phys. Rev. D **108**, 085026 (2023); [arXiv:2309.02941](#).
- [39] S. J. Summers and R. F. Werner, *Maximal violation of Bell’s inequalities is generic in quantum field theory*, Commun. Math. Phys. **110**, 247 (1987).
- [40] Z. Ji, A. Natarajan, T. Vidick, J. Wright, and H. Yuen, *MIP*=RE*, [arXiv:2001.04383](#) (2020).
- [41] Z. Fan and H. de Garis, *Braid Matrices and Quantum Gates for Ising Anyons Topological Quantum Computation*, Eur. Phys. J. B **74**, 419 (2010); [arXiv:1003.1253](#).
- [42] M. Howard and J. Vala, *Nonlocality as a Benchmark for Universal Quantum Computation in Ising Anyon Topological Quantum Computers*, Phys. Rev. A **85**, 022304 (2012); [arXiv:1112.1516](#).
- [43] S. Xu *et al.*, *Non-Abelian braiding of Fibonacci anyons with a superconducting processor*, [arXiv:2404.00091](#) (2024).
- [44] R. D. Gill, *Optimal statistical analyses of Bell experiments*, [arXiv:2209.00702](#) (2022).